

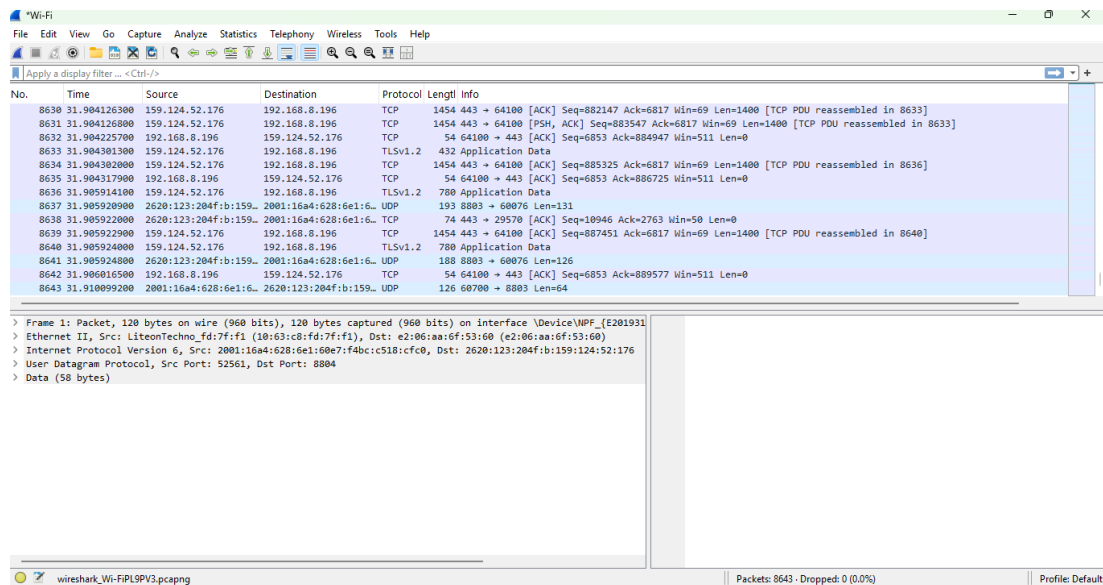
Lab 1

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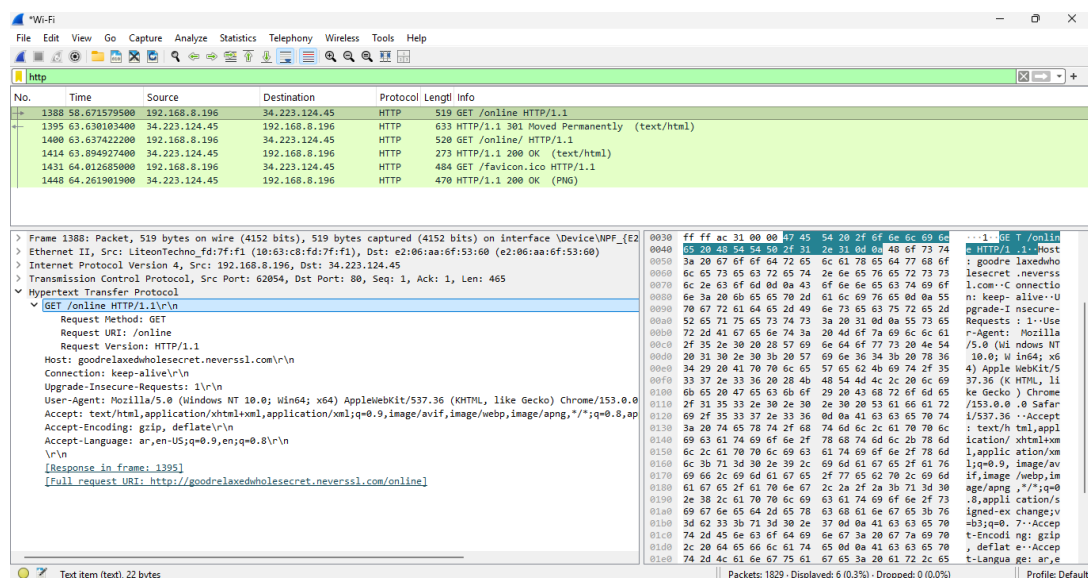
Group: 5527

Part 1: Capturing HTTP Traffic.

-Task 1: Start Wireshark and capture packets.



-Task 2: Filter HTTP packets and analyze them.



Part 2: Analyzing TCP/IP Traffic.

-Task 1: Filter TCP packets

The screenshot shows Wireshark with a filter of `tcp.stream eq 16`. The packet list shows several packets, with packet 6800 selected. The details pane shows the following information:

- Frame 6800: Packet, 669 bytes on wire (5352 bits)
- Ethernet II, Src: e2:06:aa:6f:53:60 (e2:06:aa:6f:53:60), Dst: 2600:1f13:37c:1400::200
- Internet Protocol Version 6, Src: 2600:1f13:37c:1400::200, Dst: 2600:1f13:37c:1400::200
- Transmission Control Protocol, Src Port: 80, Dst Port: 443
- Hypertext Transfer Protocol
 - HTTP/1.1 301 Moved Permanently
 - Response Version: HTTP/1.1
 - Status Code: 301
 - [Status Code Description: Moved Permanently]
 - Response Phrase: Moved Permanently
 - Date: Tue, 15 Sep 2026 12:23:40 GMT
 - Server: Apache/2.4.68 (Ubuntu)
 - Location: http://goodmajestictranscendentsecret.neverssl.com/online/
 - Content-Length: 386
 - Keep-Alive: timeout=5, max=100
 - Connection: Keep-Alive
 - Content-Type: text/html; charset=iso-8859-1
 - HTML body:

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<title>301 Moved Permanently</title>
<head>
<meta http-equiv="Content-Type" content="text/html" charset="iso-8859-1">
</head>
<body>
<h1>301 Moved Permanently</h1>
<p>The document has moved <a href="http://goodmajestictranscendentsecret.neverssl.com/online/">here</a>
</p>
</body>
</html>
```

-Task 2: Analyze TCP handshake and investigate Data Transfer and Termination.

- TCP Handshake:

The screenshot shows Wireshark with a filter of `tcp.stream eq 12`. The packet list shows several packets, with packet 4631 selected. The details pane shows the following information:

- Frame 4631: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF{E201...}
- Ethernet II, Src: LiteonTechno-fd:7f:f1 (18:63:c8:fd:7f:f1), Dst: e2:06:aa:6f:53:60 (e2:06:aa:6f:53:60)
- Internet Protocol Version 6, Src: 2001:16a4:628:6e1:6::200, Dst: 2001:16a4:628:6e1:6::200
- Transmission Control Protocol, Src Port: 63143, Dst Port: 443, Seq: 1, Ack: 1, Len: 0
- TCP Segment (raw): 3637655095
- [Next Sequence Number: 1 (relative sequence number)]
- Acknowledgment Number: 1 (relative ack number)

- Data Transfer:

The image shows a Wireshark capture of an HTTP transaction. The packet list on the left shows several packets, with packet 6716 selected. The packet details pane shows the structure of the HTTP request and response. The request is a GET for /online HTTP/1.1. The response is a 301 Moved Permanently status with a Location header pointing to http://goodmajestictranscendentsecret.neverssl.com/online/. The packet bytes pane shows the raw data of the response, including the HTML body which contains a message about the document being moved.

Wireshark - Follow TCP Stream (tcp.stream eq 16) - Wi-Fi

GET /online HTTP/1.1
Host: goodmajestictranscendentsecret.neverssl.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding: gzip, deflate
Accept-Language: ar,en-US;q=0.9,en;q=0.8

HTTP/1.1 301 Moved Permanently
Date: Tue, 15 Sep 2026 12:23:40 GMT
Server: Apache/2.4.68 ()
Location: http://goodmajestictranscendentsecret.neverssl.com/online/
Content-Length: 306
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>301 Moved Permanently</title>
</head><body>
<h1>Moved Permanently</h1>
<p>The document has moved here
</body></html>

- Termination:

The image shows a Wireshark capture of a TCP termination sequence. The packet list on the left shows several packets, with packet 4988 selected. The packet details pane shows the structure of the TCP reset (RST) packet. The packet bytes pane shows the raw data of the reset packet, including the sequence number and the reset flag.

Wireshark - Follow TCP Stream (tcp.stream eq 18) - Wi-Fi

4988 17.64454200 2600:1f13:37c:1400::... 2600:1f13:37c:1400::... TCP 74 443 → 26472 [ACK] Seq=1566 Ack=2016 Win=32768 Len=0 [TCP PDU reassembled in 4997]

4997 17.64774280 2600:1f13:37c:1400::... 2600:1f13:37c:1400::... TCP 1474 443 → 26472 [ACK] Seq=1566 Ack=2016 Win=32768 Len=1400 [TCP PDU reassembled in 4997]

5005 17.92105980 2600:1f13:37c:1400::... 2600:1f13:37c:1400::... TCP 74 443 → 26472 [ACK] Seq=4335 Ack=2080 Win=32768 Len=0

5097 17.96692960 2600:1f13:37c:1400::... 2600:1f13:37c:1400::... TCP 74 443 → 26472 [ACK] Seq=4514 Ack=2795 Win=36864 Len=0

5732 20.43754610 2600:1f13:37c:1400::... 2600:1f13:37c:1400::... TCP 74 443 → 26472 [ACK] Seq=4514 Ack=2796 Win=36864 Len=0

4988 17.64454200 2600:1f13:37c:1400::... 2600:1f13:37c:1400::... TCP 74 443 → 26472 [RST] Seq=0 Win=0 Len=0 MSS=1440 WS=256 SACK_PERM

Source Port: 443
Destination Port: 26472
[Stream index: 18]
[Stream Packet Number: 6]
[Conversation completeness: Complete, WITH_DATA (63)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 71316224
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 2016 (relative ack number)
Acknowledgment Number (raw): 4136173879

Part 3: Capturing and Analyzing UDP Traffic

- Task 1: Generate UDP traffic and capture packets

```
Command Prompt

Non-authoritative answer:
Name:    google.com
Addresses: 2a00:1450:4006:81d::200e
          142.250.181.206

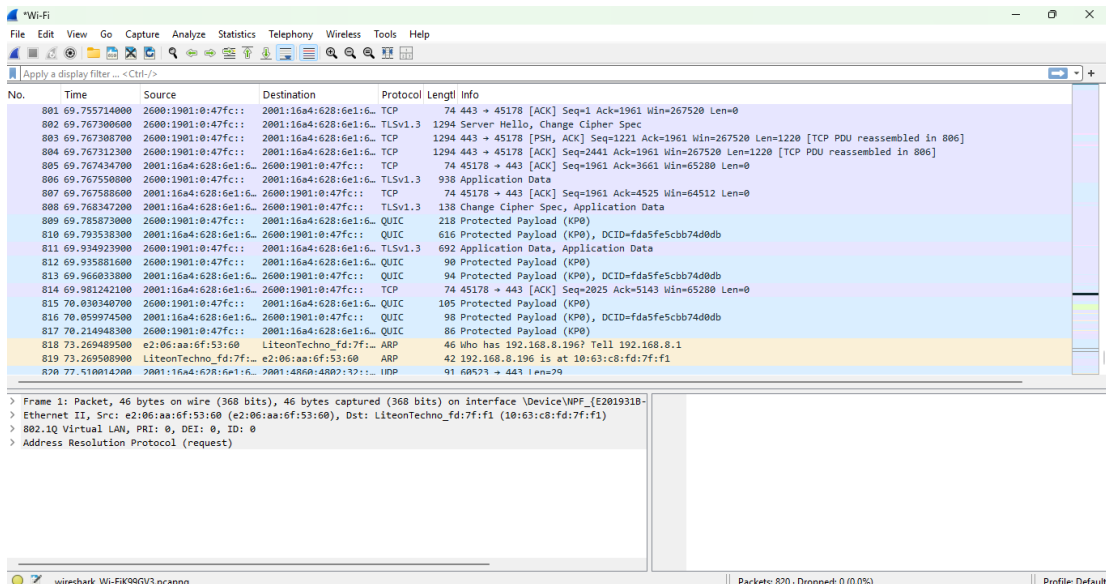
C:\Users\occ>nslookup youtube.com
Server:    homerouter.cpe
Address:    fe80::1be7:e4d2:d073:65c8

Non-authoritative answer:
Name:    youtube.com
Addresses: 2a00:1450:4006:817::200e
          172.217.18.110

C:\Users\occ>nslookup example.com
Server:    homerouter.cpe
Address:    fe80::1be7:e4d2:d073:65c8

Non-authoritative answer:
Name:    example.com
Addresses: 2606:4700:10::ac42:93f3
          2606:4700:10::6814:179a
          104.20.23.154
          172.66.147.243

C:\Users\occ>
```



- Task 2: Filter and analysis UDP Packets

UDP is simpler than TCP because it has only four main fields and does not need to manage connection details like TCP. TCP requires additional fields to ensure reliable and ordered data delivery.

The screenshot shows a Wireshark packet capture of UDP traffic. The packet list at the top shows several UDP packets. The packet details pane for packet 89 shows the following information:

- Frame 89: Packet, 131 bytes on wire (1048 bits), 131 bytes captured (1048 bits) on interface \Device\NPF_{E201...}
- Ethernet II, Src: e2:06:aa:f5:53:60 (e2:06:aa:f5:53:60), Dst: LiteonTechno_fd:7f:f1 (10:63:c8:fd:7f:f1)
- Internet Protocol Version 6, Src: 2001:16a4:628:6e1:6, Dst: 2001:16a4:628:6e1:6
- User Datagram Protocol, Src Port: 443, Dst Port: 60523
- Source Port: 443
- Destination Port: 60523
- Length: 77
- Checksum: 0x7568 [unverified]
- [Checksum Status: Unverified]
- [Stream index: 0]
- [Stream Packet Number: 1]
- [Timestamps]
- UDP payload (69 bytes)
- Data (69 bytes)

The packet bytes pane shows the raw data of the UDP packet, including the header and payload.

Part 4: Comparing TCP and UDP by filling in the following tables. Save your work (e.g., in an MS Word document), and upload it to your online git repo.

-Task 1: Fill in the following table and provide reasons.

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP because it uses the Three-Way Handshake before transferring data and uses ACKs to confirm that the data has been received. It can also retransmit lost data to ensure reliable delivery.
Data Integrity and Ordering	TCP	TCP because it ensures data integrity by retransmitting lost data and ensures that data is delivered in the correct order using sequence numbers.

- Task 2: Identify the use Cases and Performance of TCP and UDP.

	TCP	UDP
Use cases	• Web Browsing – HTTP/HTTPS • Email – SMTP • Remote Access – SSH	• Online Games • Voice over IP (VoIP) • Domain Name System (DNS)
Performance	TCP is generally slower than UDP because it has more fields and performs additional tasks to ensure reliable and ordered data delivery.	UDP is generally faster than TCP because it does not establish a connection or ensure reliable delivery of data, so packets may be lost.